

PHYS 230 - Electricity and Magnetism

Lecture 32

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

8 Sources of Magnetic Field

- Force Between Parallel Conductors (Continued)
- Magnetic Field of a Circular Current Loop

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

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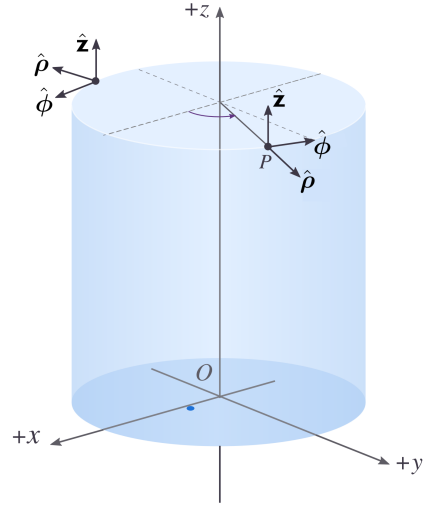
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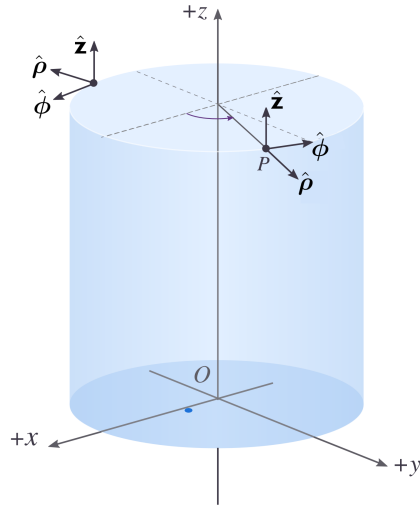
Cylindrical Coordinates

- $\hat{\mathbf{z}}$ is the unit vector along z axis,



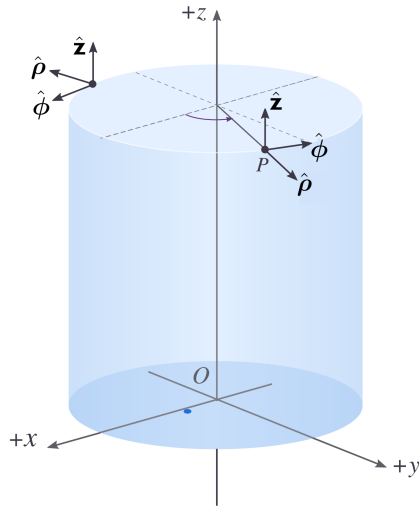
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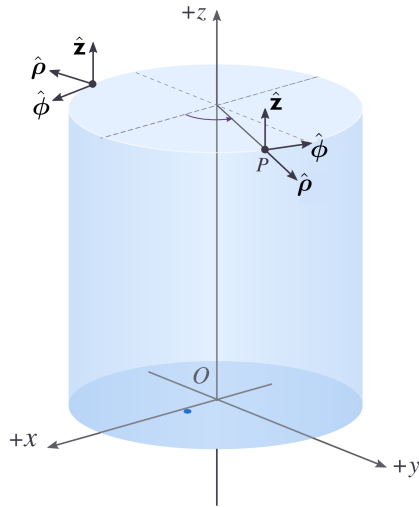
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 - Thumb of the right hand in the direction of $\hat{\mathbf{z}}$, then rotation of four other fingers will be in the direction of $\hat{\phi}$



Force Between Two Long Wires

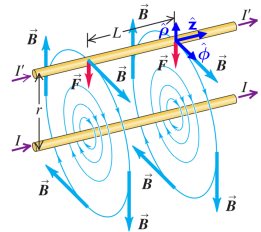
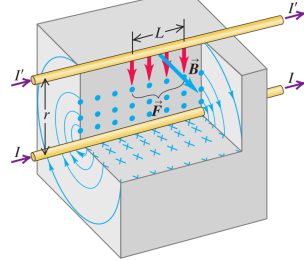
Consider two straight current-carrying conductors, one with current I and another with I' . The force on a length L of the wire carrying I' by the wire carrying I is (force from I on I')

$$\mathbf{F}_{I'} = I' \mathbf{L} \times \mathbf{B}_I$$

where

The magnetic field of the lower wire exerts an attractive force on the upper wire. By the same token, the upper wire attracts the lower one.

If the wires had currents in *opposite* directions, they would *repel* each other.



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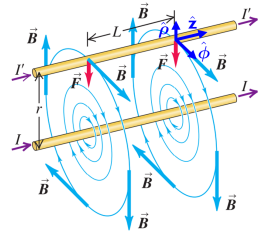
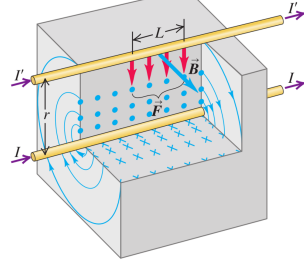
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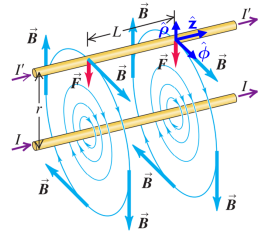
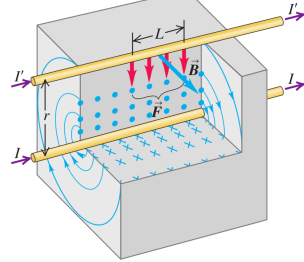
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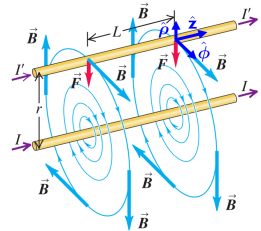
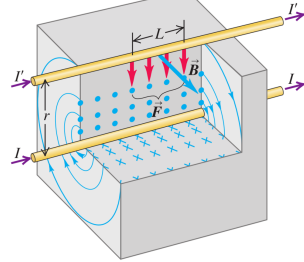
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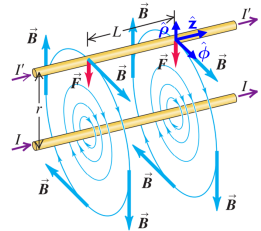
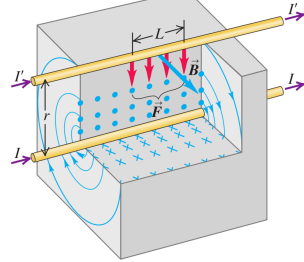
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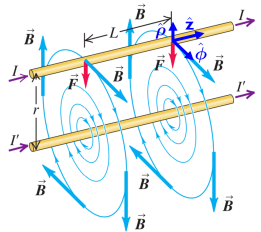
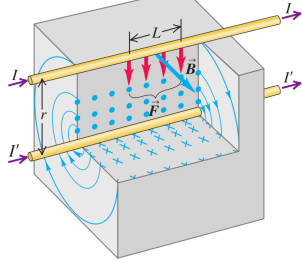
If we consider the force $\mathbf{F}_{l'l}$ from l' on l we get exactly the same expression but then $\hat{\rho}$ would be the radial unit vector from l' to l

$$\mathbf{F}_{l'l} = I\mathbf{L} \times \mathbf{B}_{l'}$$

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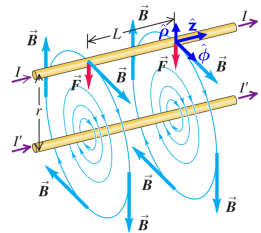
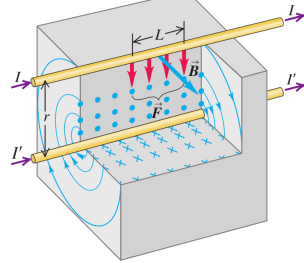
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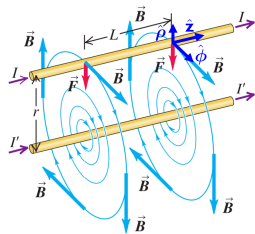
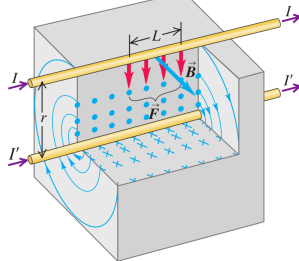
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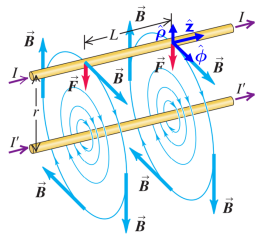
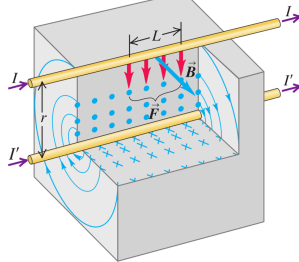
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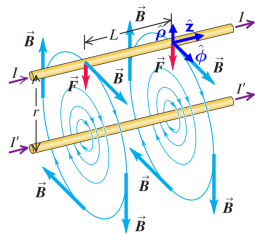
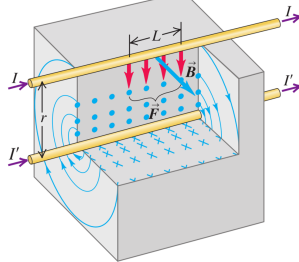
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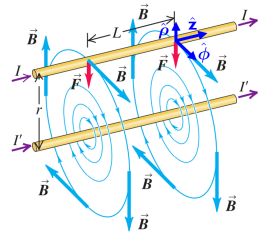
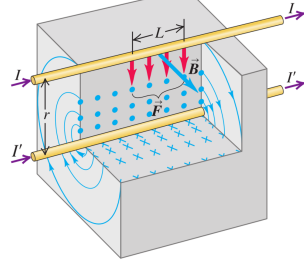
Force Between Two Long Wires

So both $\mathbf{F}_{II'}$ and $\mathbf{F}_{I'I}$ are **attractive** with a magnitude of

$$|\mathbf{F}_{II'}| = |\mathbf{F}_{I'I}| = F = \frac{\mu_0}{2\pi} \frac{II'L}{r}$$

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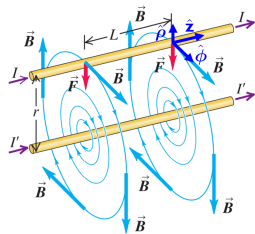
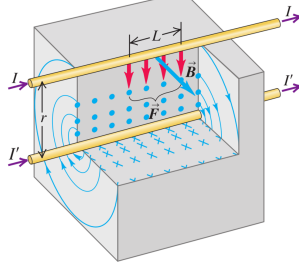
$$|\mathbf{F}_{II'}| = |\mathbf{F}_{I'I}| = F = \frac{\mu_0}{2\pi} \frac{II'L}{r}$$

The magnitude of this force per unit length of wires is

$$\frac{F}{L} = \frac{\mu_0}{2\pi} \frac{II'}{r}$$

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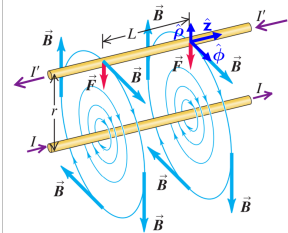
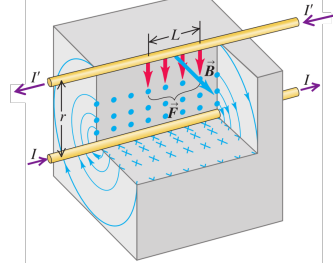
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If one of the currents is reversed, e.g., I' goes in the $-\hat{z}$ direction, the force $\mathbf{F}_{I'}$ from I on I' becomes

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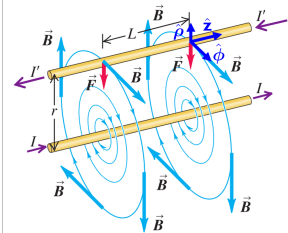
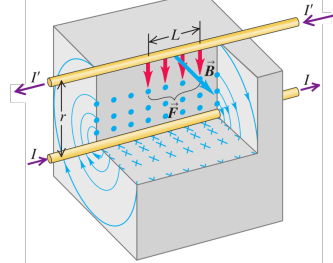
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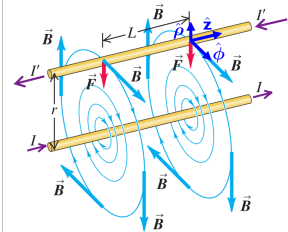
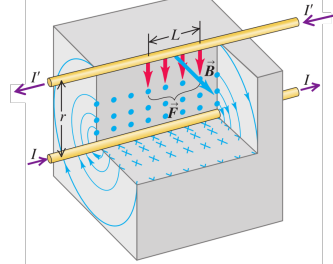
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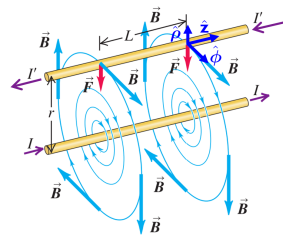
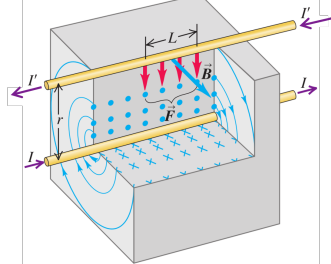
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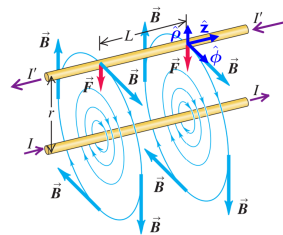
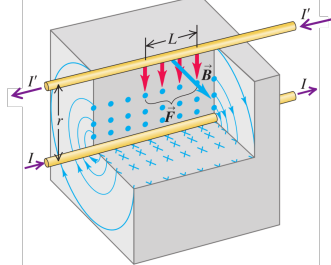
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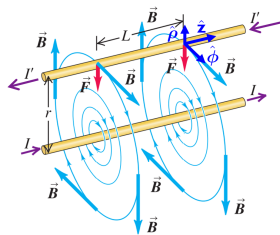
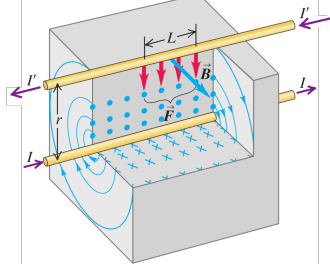
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Now the force are **repulsive!** but they have the same magnitude as before

$$F = \frac{\mu_0}{2\pi} \frac{II' L}{r}$$

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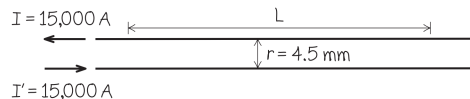
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Exercise 1

<https://epoll.srv.ualberta.ca> – Code: MBD

Two straight, parallel, superconducting wires 4.5 mm apart carry equal currents of 15000 A in opposite directions. What force, per unit length, does each wire exert on the other?



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- Magnetic Field of a Circular Current Loop

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

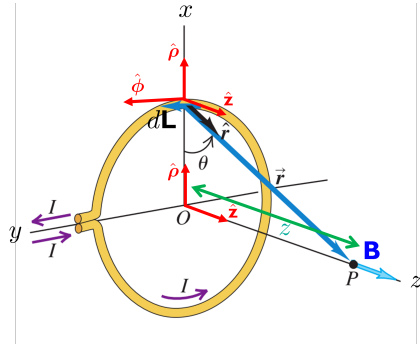
Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Magnetic Field at a Point On the Axis of a Loop

At any point P on the axis of a current carrying loop of current I , the magnetic field $d\mathbf{B}$ due to a small element $d\mathbf{L}$ of the loop is

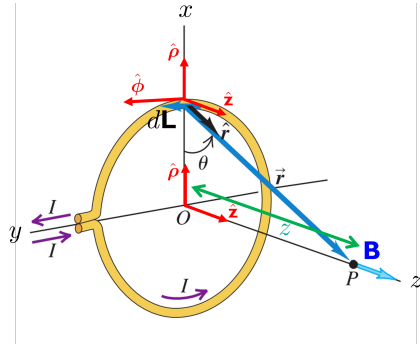
$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \mathbf{r}}{r^3}$$



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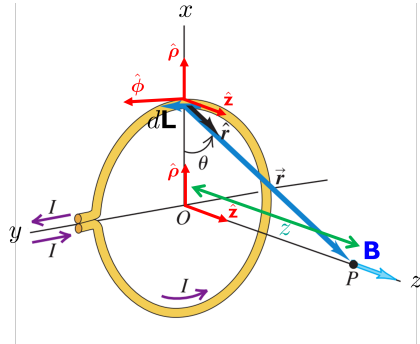
$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{I (dL \hat{\phi}) \times [R(-\hat{\rho}) + z\hat{z}]}{r^3}$$



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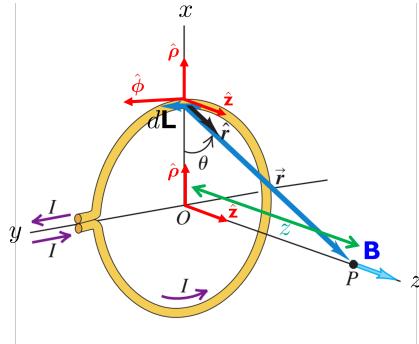
$$\begin{aligned} d\mathbf{B} &= \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{I (dL \hat{\phi}) \times [R(-\hat{\rho}) + z\hat{\mathbf{z}}]}{r^3} \\ &= \frac{\mu_0}{4\pi} \frac{IdL [-R\hat{\phi} \times \hat{\rho} + z\hat{\phi} \times \hat{\mathbf{z}}]}{r^3} \end{aligned}$$



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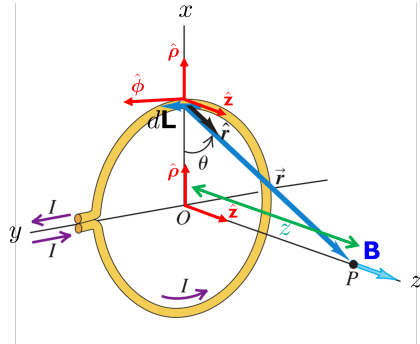
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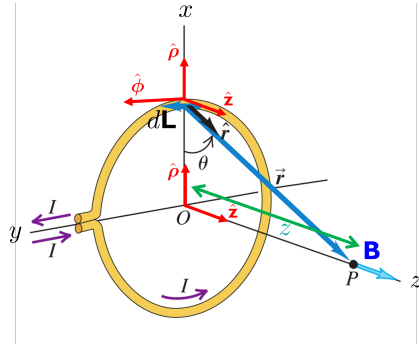
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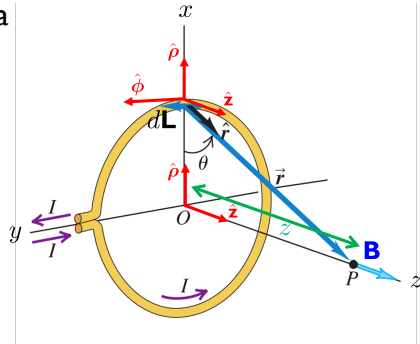
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Integrating this from $\phi = 0$ to $\phi = 2\pi$ we obtain the magnetic field at a point P on the z axis (the axis of the loop) as

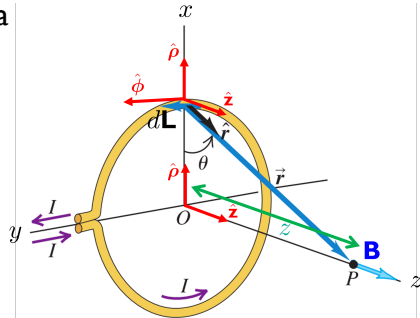
$$\mathbf{B} = \int d\mathbf{B} = \int_0^{2\pi} \frac{\mu_0}{4\pi} \frac{IR [R\hat{\mathbf{z}} + z\hat{\boldsymbol{\rho}}]}{r^3} d\phi$$



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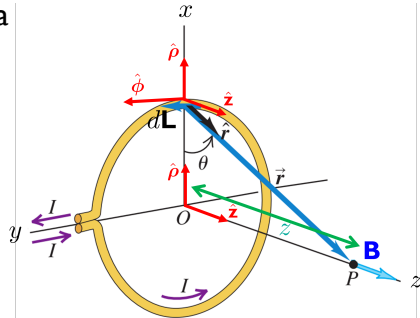
$$\begin{aligned}\mathbf{B} &= \int d\mathbf{B} = \int_0^{2\pi} \frac{\mu_0 IR}{4\pi} \frac{[R\hat{\mathbf{z}} + z\hat{\boldsymbol{\rho}}]}{r^3} d\phi \\ &= \frac{\mu_0 IR}{4\pi r^3} \left[\int_0^{2\pi} R\hat{\mathbf{z}} d\phi + \int_0^{2\pi} z\hat{\boldsymbol{\rho}} d\phi \right]\end{aligned}$$



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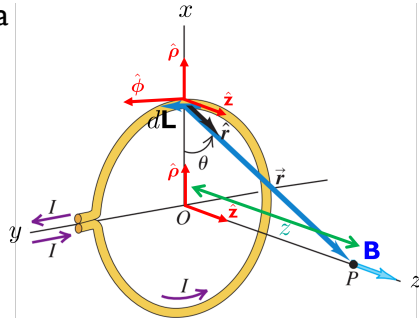
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We have $\int_0^{2\pi} \hat{\boldsymbol{\rho}} d\phi = 0$ because after a 2π rotation, $\hat{\boldsymbol{\rho}}$ turns into itself, so

$$\mathbf{B} = \frac{\mu_0 IR^2}{4\pi r^3} \left[\int_0^{2\pi} d\phi \right] \hat{\mathbf{z}}$$



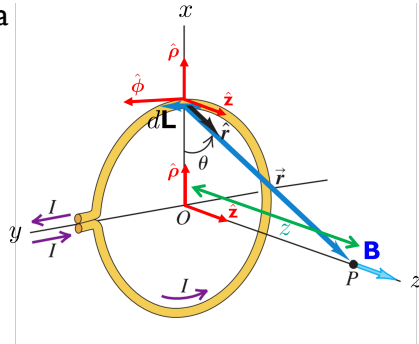
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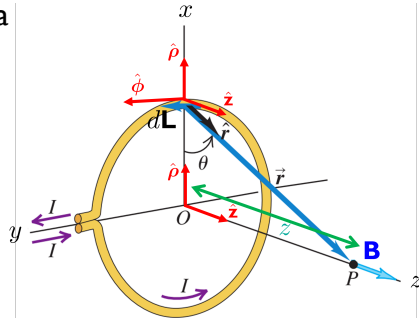
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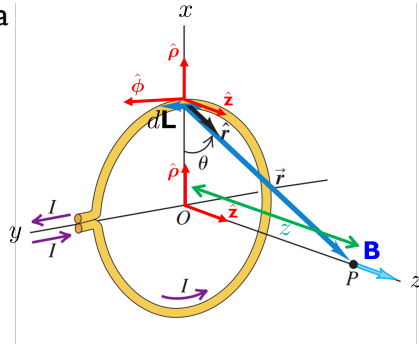
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Note that the direction of $\mathbf{B}$ (in this case $\hat{\mathbf{z}}$) is still given by the right hand rule!

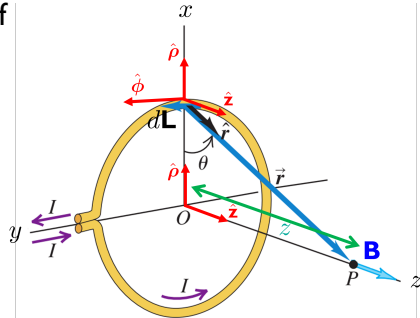


Magnetic Field at a Point On the Axis of a Loop

So the magnetic field of a loop of counterclockwise (in the direction of $\hat{\phi}$) current I with radius R , on the axis of the loop and at a distance z from the center of the loop is

$$\mathbf{B} = \frac{\mu_0 I}{2} \frac{R^2}{(z^2 + R^2)^{\frac{3}{2}}} \hat{\mathbf{z}}$$

for a clockwise current (in the direction of $-\hat{\phi}$) we get a negative in the above (in the $-\mathbf{z}$ direction)



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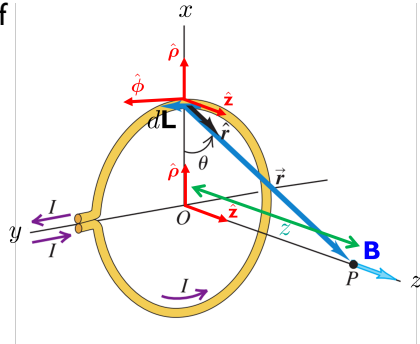
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At the center of the loop where $z = 0$ we get

$$\mathbf{B} = \frac{\mu_0 I}{2R} \hat{\mathbf{z}}$$

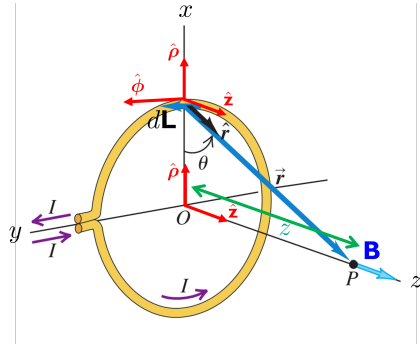


Magnetic Field at a Point On the Axis of a Coil

If we have a coil of N loops, the magnetic field of a counterclockwise current I with coil radius R , on the axis of the coil and at a distance z from the center of the coil is

$$\mathbf{B} = \frac{\mu_0 N I}{2} \frac{R^2}{(z^2 + R^2)^{\frac{3}{2}}} \hat{\mathbf{z}}$$

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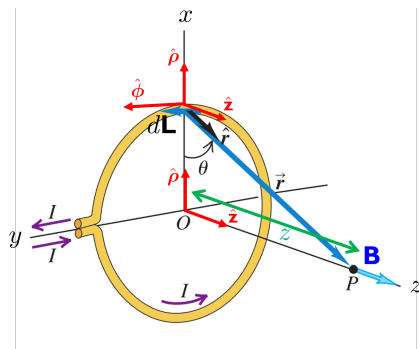
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Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD

A coil consisting of 100 circular loops with radius 0.60 m carries a 5.0 A current. (a) Find the magnitude of the magnetic field at a point along the axis of the coil, 0.80 m from the center. (b) Along the axis, at what distance from the center of the coil is the field magnitude $\frac{1}{8}$ as great as it is at the center?

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